$$\frac{1+v}{v^2+2v-1}dv = \frac{-dx}{x}$$

Multiply 2 on both sides

$$\frac{2+2v}{v^2+2v-1}dv = -2\frac{dx}{x}$$

On Integration

$$\int\frac{2+2v}{v^2+2v-1}dv = -2\int\frac{dx}{x}$$

$$\log(v^2+2v-1) = -2\log x + \log c$$

$$v^2+2v-1 = \frac{c}{x^2}$$

$$x^2\,(v^2+2v-1) = c$$

Now, Replace $v = \frac{y}{x}$

$$x^2\left[\frac{y^2}{x^2}+\frac{2y}{x}-1\right] = c$$

$$y^2+2xy-x^2 = c \text{ is the solution.}$$

Example 4.17

Find the particular solution of the differential equation $x^2\,dy + y\,(x+y)\,dx = 0$ given that $x=1,\ y=1$

Solution

$$x^2dy + y(x+y)dx = 0$$

$$x^2dy = -y\,(x+y)\,dx$$

$$\frac{dy}{dx} = \frac{-(xy+y^2)}{x^2} \quad (1)$$

Put $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ in (1)

$$v + x\frac{dv}{dx} = \frac{-(xvx+v^2x^2)}{x^2}$$

$$= -\left(v+v^2\right)$$

$$x\frac{dv}{dx} = -v^2-v-v$$

$$= -(v^2+2v)$$

On separating the variables

$$\frac{dv}{v^2+2v} = \frac{-dx}{x}$$

$$\frac{dv}{v(v+2)} = \frac{-dx}{x}$$

$$\frac{1}{2}\left[\frac{(v+2)-v}{v(v+2)}\right]dv = \frac{-dx}{x}$$

$$\frac{1}{2}\int\left(\frac{1}{v}-\frac{1}{v+2}\right)dv = -\int\frac{dx}{x}$$

$$\frac{1}{2}\left[\log v - \log(v+2)\right] = -\log x + \log c$$

$$\frac{1}{2}\log\frac{v}{v+2} = \log\frac{c}{x}$$

We have

$$\frac{v}{v+2} = \frac{c^2}{x^2}$$

Replace $v = \frac{y}{x}$, we get

$$\frac{\frac{y}{x}}{x\left(\frac{y}{x}+2\right)} = \frac{k}{x^2}$$

where $c^2 = k$

$$\frac{y\,x^2}{y+2x} = k \quad (2)$$

When $x=1,\ y=1$

$$\therefore (2) \Rightarrow k = \frac{1}{1+2}$$

$$k = \frac{1}{3}$$

$\therefore$ The solution is $3x^2\,y = 2x+y$

Example 4.18

If the marginal cost of producing x shoes is given by $(3xy+y^2)\,dx + (x^2+xy)\,dy = 0$. and the total cost of producing a pair of shoes is given by ₹12. Then find the total cost function.

Solution:

Given marginal cost function is $(x^2+xy)\,dy + (3xy+y^2)\,dx = 0$

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